Phys 115B HW 1

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Question 1. ***You just know the answer***
A common problem in QM is "find the energies of the following Hamiltonian." The idea is to do as little math as possible but to build intuition. Describe the lowest several eigenenergies of the following Hamiltonians. It is up to you to decide what value of "several" best illustrates what is going on. Answers are short.
Example: *What are the energies of the two separate simple harmonic oscillators, each with frequency ω?*
Answer: *The oscillators have energies*

$$E_1 = \hbar\omega(1/2 + n_1), \quad n_1 = [0, 1, 2, 3, \ldots]$$

$$E_2 = \hbar\omega(1/2 + n_2), \quad n_2 = [0, 1, 2, 3, \ldots]$$

so the combined oscillators have energy

$$E = E_1 + E_2 = \hbar\omega(1 + n_1 + n_2) = \hbar\omega[1, 2, 2, 3, 3, 3, 4, 4, 4, 4, \ldots]$$

1. *Particle of mass m in one dimensional box of width a.*
2. *Particle of mass m in two dimensional box of width a and length b.*
3. *Two disconnected particles, each of mass m, one in a 1D box of width a and the other in a 1D box of width b.*
4. *Particle of mass m confined to a circular ring of circumference a.*
5. *Particle of mass m in a one dimensional potential $V = \frac{1}{2}m\omega^2x^2$.*
6. *Paricle of mass m in a two dimensional potential $V = \frac{1}{2}m\omega^2(x^2 + y^2)$.*

Proof.

1. Since the partical is limited in the box (and can assume it's impossible to be outside of the box), it's identical to an infinite square well with width a. Which the first three energies are given by $E_1 = \frac{\pi^2\hbar^2}{2ma^2}, E_2 = \frac{4\pi^2\hbar^2}{2ma^2}$, and $E_3 = \frac{9\pi^2\hbar^2}{2ma^2}$.
2. Similar to part (a), the particle is an infinite square well of width a in one direction, while an infinite square well of width b in the other direction. Hence, the energies are the sum of the two infinite square

wells (since the x, y directions are independent of each other, which the energies are also independent), which has the first several energies $E_{1,1} = \frac{\pi^2\hbar^2}{2ma^2} + \frac{\pi^2\hbar^2}{2mb^2}$, $E_{1,2} = \frac{\pi^2\hbar^2}{2ma^2} + \frac{4\pi^2\hbar^2}{2mb^2}$, $E_{2,1} = \frac{4\pi^2\hbar^2}{2ma^2} + \frac{\pi^2\hbar^2}{2mb^2}$, and $E_{2,2} = \frac{4\pi^2\hbar^2}{2ma^2} + \frac{4\pi^2\hbar^2}{2mb^2}$.

3. Since the two particles are disconnected, the total energy is the sum of the two particles' energy. Since both particles are within an infinite square well (with width a, b respectively), then the energies are similar to part 2: $E_{1,1} = \frac{\pi^2\hbar^2}{2ma^2} + \frac{\pi^2\hbar^2}{2mb^2}$, $E_{1,2} = \frac{\pi^2\hbar^2}{2ma^2} + \frac{4\pi^2\hbar^2}{2mb^2}$, $E_{2,1} = \frac{4\pi^2\hbar^2}{2ma^2} + \frac{\pi^2\hbar^2}{2mb^2}$, and $E_{2,2} = \frac{4\pi^2\hbar^2}{2ma^2} + \frac{4\pi^2\hbar^2}{2mb^2}$.

4. The particle is limited to a circular ring of circumference a, which is similar to an infinite square well with width a (except now its derivative needs to be continuous, and agrees on both 0 and a). Hence, here the odd energies cannot work (since the solution where the spatial frequency of the solution $\frac{\sqrt{2mE}}{\hbar} = \frac{n\pi}{a}$ only completes a full cycle, or having the derivatives agree on the boundary when n is even). Hence, the first several energies are $E_2 = \frac{4\pi^2\hbar^2}{2ma^2}$, $E_4 = \frac{16\pi^2\hbar^2}{2ma^2}$, and $E_6 = \frac{36\pi^2\hbar^2}{2ma^2}$.

5. Since the potential is a one-dimensional simple harmonic oscillator, the first several energies are given by $E_0 = \frac{\hbar w}{2}$, $E_1 = \frac{3\hbar w}{2}$, $E_2 = \frac{5\hbar w}{2}$, and $E_3 = \frac{7\hbar w}{2}$.

6. With the potential being a two-dimensional harmonic oscillator (with both the x, y direction having the same frequency ω), then the energies are the sum of two one-dimensional harmonic oscillators (both have frequency w). Hence, the first several energies are $E_{0,0} = \frac{\hbar w}{2} + \frac{\hbar w}{2} = \hbar w$, $E_{0,1} = E_{1,0} = \frac{\hbar w}{2} + \frac{3\hbar w}{2} = 2\hbar w$, and $E_{1,1} = \frac{3\hbar w}{2} + \frac{3\hbar w}{2} = 3\hbar w$.

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Question 2. ***Two-body to one-body*** *(Griffiths 5.1 in both editions)*
Typically, the interaction potential depends onlly on the vector $\mathbf{r} := \mathbf{r}_1 - \mathbf{r}_2$ *separating the two particles. In that case the Schrödinger equation separates, if we change variables from* $\mathbf{r}_1, \mathbf{r}_2$ *to* $\mathbf{r}$, $\mathbf{R} := (m_1\mathbf{r}_1 + m_2\mathbf{r}_2)/(m_1+m_2)$ *(the center of mass).*

(a) *Show that* $\mathbf{r}_1 = \mathbf{R} + (\mu/m_1)\mathbf{r}$, $\mathbf{r}_2 = \mathbf{R} - (\mu/m_2)\mathbf{r}$, *and* $\nabla_1 = (\mu/m_2)\nabla_R + \nabla_r$, $\nabla_2 = (\mu/m_1)\nabla_R - \nabla_r$, *where*

$$\mu := \frac{m_1 m_2}{m_1 + m_2}$$

is the ***reduced mass*** *of the system.*

(b) *Show that the (time-independent) Schrödinger equation becomes*

$$-\frac{\hbar^2}{2(m_1+m_2)}\nabla_R^2\psi - \frac{\hbar^2}{2\mu}\nabla_r^2\psi + V(\mathbf{r})\psi = E\psi$$

Note: the original equation is:

$$-\frac{\hbar^2}{2m_1}\nabla_1^2\psi - \frac{\hbar^2}{2m_2}\nabla_2^2\psi + V\psi = E\psi$$

(c) *Separate the variables, letting* $\psi(\mathbf{R},\mathbf{r}) = \psi_R(\mathbf{R})\psi_r(\mathbf{r})$. *Note that* ψ_R *satisfies the one-particle Schrödiger equation, with the* total *mass* (m_1+m_2) *in place of* m, *potential zero, and energy* E_R, *while* ψ_r *satisfies the one-particle Schrödinger equation with the* reduced *mass in place of* m, *potential* $V(\mathbf{r})$, *and energy* E_r. *The total energy is the sum:* $E = E_R + E_r$. *What this tells us is that the center of mass moves like a free particle, and the* relative motion *(that is, the motion of particle 2 with respect to particle 1) is the same as if we had a* single *particle with the* reduced *mass, subject to the potential* V. *Exactly the same decomposition occurs in* classical mechanics*; it reduces the two-body problem to an equivalent one-body problem.*

Proof.

(a) First, about $\mathbf{r}_1, \mathbf{r}_2$, one has the following:

$$\mathbf{R} + \frac{\mu}{m_1}\mathbf{r} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1+m_2} + \frac{m_2}{m_1+m_2}(\mathbf{r}_1 - \mathbf{r}_2) = \frac{(m_1+m_2)\mathbf{r}_1}{m_1+m_2} = \mathbf{r}_1 \tag{1}$$

$$\mathbf{R} - \frac{\mu}{m_2}\mathbf{r} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1+m_2} - \frac{m_1}{m_1+m_2}(\mathbf{r}_1 - \mathbf{r}_2) = \frac{(m_1+m_2)\mathbf{r}_2}{m_1+m_2} = \mathbf{r}_2 \tag{2}$$

Now, let $\{x_1, y_1, z_1\}$ and $\{x_2, y_2, z_2\}$ denote the coordinates of $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively. Then, $\mathbf{r}$ has coordinates $\{x_1 - x_2, y_1 - y_2, z_1 - z_2\}$, while $\mathbf{R}$ has coordinates $\{X, Y, Z\} = \{\frac{m_1x_1+m_2x_2}{m_1+m_2}, \frac{m_1y_1+m_2y_2}{m_1+m_2}, \frac{m_1z_1+m_2z_2}{m_1+m_2}\}$. So, given $u \in \{x, y, z\}$, the transformation $(u_1, u_2) = f(U, u)$ can be written as follow:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} U + \frac{\mu}{m_1}u \\ U - \frac{\mu}{m_2}u \end{pmatrix} = \begin{pmatrix} 1 & \frac{\mu}{m_1} \\ 1 & -\frac{\mu}{m_2} \end{pmatrix}\begin{pmatrix} U \\ u \end{pmatrix} \tag{3}$$

As a result, the differential operators with respect to U, u are given as follow:

$$\frac{\partial}{\partial U} = \frac{\partial u_1}{\partial U}\frac{\partial}{\partial u_1} + \frac{\partial u_2}{\partial U}\frac{\partial}{\partial u_2} = \frac{\partial}{\partial u_1} + \frac{\partial}{\partial u_2} \tag{4}$$

$$\frac{\partial}{\partial u} = \frac{\partial u_1}{\partial u}\frac{\partial}{\partial u_1} + \frac{\partial u_2}{\partial u}\frac{\partial}{\partial u_2} = \frac{\mu}{m_1}\frac{\partial}{\partial u_1} - \frac{\mu}{m_2}\frac{\partial}{\partial u_2} \tag{5}$$

So, the square differential operators become:

$$\frac{\partial^2}{\partial U^2} = \frac{\partial^2}{\partial u_1^2} + \frac{\partial^2}{\partial u_2^2} + 2\frac{\partial^2}{\partial u_1 \partial u_2} \tag{6}$$

$$\frac{\partial^2}{\partial u^2} = \frac{\mu^2}{m_1^2}\frac{\partial^2}{\partial u_1^2} + \frac{\mu^2}{m_2^2}\frac{\partial^2}{\partial u_2^2} - \frac{2\mu^2}{m_1 m_2}\frac{\partial^2}{\partial u_1 \partial u_2} \tag{7}$$

$$= \frac{1}{(m_1+m_2)^2}\left(m_2^2\frac{\partial^2}{\partial u_1^2} + m_1^2\frac{\partial^2}{\partial u_2^2} - 2m_1 m_2\frac{\partial^2}{\partial u_1 \partial u_2}\right) \tag{8}$$

Which, the given

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Question 3. ***SHO in 3D*** *(Modified Griffiths 3rd ed. 4.46, or 2nd ed. 4.39)*
Consider a particle of mass m for which the potential is

$$V(r) = \frac{1}{2}m\omega^2 r^2$$

(a) Using separation of variables in Cartesian coordinates, show that this factorizes into a sum of three 1D harmonic oscillators, and use your knowledge of 1D SHO to determine the allowed energies.

(b) Determine the degeneracy $d(n)$ (i.e. the number of states with the same energy as the nth energy eigenvale E_n.

(c) As you hopefully found in part (b), the ground state $\psi_0(x, y, z)$ is non-degenerate. However the first excited energy eigenvalue E_1 is threefold degenerate, with eigen- states:

$$\psi_{1,0,0}(x, y, z) = \tilde{\psi}_1(x)\tilde{\psi}_0(y)\tilde{\psi}_0(z)$$

$$\psi_{0,1,0}(x, y, z) = \tilde{\psi}_0(x)\tilde{\psi}_1(y)\tilde{\psi}_0(z)$$

$$\psi_{0,0,1}(x, y, z) = \tilde{\psi}_0(x)\tilde{\psi}_0(y)\tilde{\psi}_1(z)$$

where $\tilde{\psi}_i$ denotes an eigenstate of the 1D harmonic oscillator. Are any of the linear combinations

$$\frac{1}{\sqrt{3}}(\psi_{1,0,0} + \psi_{0,1,0} + \psi_{0,0,1}), \quad \frac{1}{\sqrt{2}}(\psi_{1,0,0} + \psi_{0,1,0}), \quad \frac{1}{\sqrt{2}}(\psi_{1,0,0} + i\psi_{0,0,1})$$

also energy eigenstates? In each case, why or why not?

Proof.

(a) Suppose $\psi(\mathbf{r}) = \psi_x(x)\psi_y(y)\psi_z(z)$, then plugin to the Time-independent Schrödinger's Equation, we get:

$$\begin{aligned} E\psi_x\psi_y\psi_z &= -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_x\psi_y\psi_z + \frac{1}{2}mw^2(x^2+y^2+z^2)\psi_x\psi_y\psi_z \qquad (9) \\ &= -\frac{\hbar^2}{2m}\left(\psi_y\psi_z\frac{\partial^2\psi_x}{\partial x^2} + \psi_x\psi_z\frac{\partial^2\psi_y}{\partial y^2} + \psi_x\psi_y\frac{\partial^2\psi_z}{\partial z^2}\right) + \frac{1}{2}mw^2(x^2+y^2+z^2)\psi_x\psi_y\psi_z \qquad (10) \end{aligned}$$

On some open subset of $\mathbb{R}^3$ (since $\psi \neq 0$ and it's continuous), then after dividing everything by ψ, one gets:

$$E = \left(-\frac{\hbar^2}{2m\psi_x}\frac{\partial^2\psi_x}{\partial x^2} + \frac{1}{2}mw^2x^2\right) + \left(-\frac{\hbar^2}{2m\psi_y}\frac{\partial^2\psi_y}{\partial y^2} + \frac{1}{2}mw^2y^2\right) + \left(-\frac{\hbar^2}{2m\psi_z}\frac{\partial^2\psi_z}{\partial z^2} + \frac{1}{2}mw^2z^2\right) \qquad (11)$$

Which, each of the term is ony dependent of x,y, or z (but not the other two variables). Hence, in case for the equation to hold, all three terms in the paranthesis must necessarily be constant. Hence, for each $u \in \{x, y, z\}$, one has some $E_u \in \mathbb{R}$ that satisfies the following:

$$\left(-\frac{\hbar^2}{2m\psi_u}\frac{\partial^2\psi_u}{\partial u^2} + \frac{1}{2}mw^2u^2\right) = E_u \implies E_u\psi_u = -\frac{\hbar^2}{2m}\frac{\partial^2\psi_u}{\partial u^2} + \frac{1}{2}mw^2u^2\psi_u \qquad (12)$$

Hence, each ψ_u not only satisfies the one-dimensional harmonic oscillator Hamiltonian, and it's also normalizable (based on the fact that ψ is normalizable).

Finally, since ψ_x, ψ_y, ψ_z are all one-dimensional harmonic oscillators' solution (together with $E = E_x + E_y + E_z$ based on the above equation), then each $E_u = \hbar w \left(\frac{1}{2} + n_u\right)$ for some $n_u \in \mathbb{N}$. Hence, one has $E = \hbar w \left(\frac{3}{2} + n_x + n_y + n_z\right)$ for $n_x, n_y, n_z \in \mathbb{N}$, showing the allowed energies are $E = \hbar w \left(\frac{3}{2} + n\right)$ for all $n \in \mathbb{N}$.

(b) The degerneracy $d(n)$ for the nth energy eigenvalue $E_n = \hbar w \left(\frac{3}{2} + n\right)$, is given by the number of distinct 3-tuples $(n_x, n_y, n_z) \in \mathbb{N}^3$, such that $n = n_x + n_y + n_z$.

First, there are $(n+1)$ ways of fixing n_x (since $n_x \in \{0, 1, ..., n\}$). Then, for each $k \in \{0, 1, ..., n\}$, fixing $n_x = k$, the remaining $(n-k)$ can be partitianed in $(n-k+1)$ ways into 2 nonnegative integers (since for all $l \in \{0, 1, ..., n-k\}$, $(n-k) = l + (n-k-l)$). So, the degeneracy is the sum of all such possible partitian, which is given as follow:

$$d(n) = \sum_{k=0}^{n} (n-k+1) = (n+1)^2 - \frac{n(n+1)}{2} = \frac{(n+1)(n+2)}{2} \tag{13}$$

(c) All the given linear combinations of the energy eigenstates of energy eigenvalue E_1 is also an energy eigenstate, since they're all inside the same eigenspace corresponding to E_1, hence any linear combination still lies in the same space.

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Question 4. ***1D to 3D*** *(Griffiths 4.1, same in both editions)*

(a) *Work out all of the* ***canonical commutation relations*** *for components of the operators* $\mathbf{r}$ *and* $\mathbf{p}$*:* $[\hat{x}, \hat{y}], [\hat{x}, \hat{p}_y], [\hat{x}, \hat{p}_x], [\hat{p}_y, \hat{p}_z]$*, and so on.*

(b) *Confirm the three-dimensional version of* ***Ehrenfest's Theorem****,*

$$\frac{d}{dt}\langle \mathbf{r} \rangle = \frac{1}{m}\langle \mathbf{p} \rangle, \quad \frac{d}{dt}\langle \mathbf{p} \rangle = \langle -\nabla V \rangle$$

(c) *Formulate* ***Heisenberg's Uncertainty Principle*** *in three dimensions.*

Proof.

(a) Given $j, k \in \{x, y, z\}$, the action of $[\hat{j}, \hat{k}]$ on the states in position basis are as follow:

$$[\hat{j}, \hat{k}] |\psi\rangle = (\hat{j}\hat{k} - \hat{k}\hat{j}) \int_{\mathbb{R}^3} \psi(\hat{r}) |\bar{r}\rangle \, dr^3 = (\hat{j}k \int_{\mathbb{R}^3} k\psi(\bar{r}) |\bar{r}\rangle \, dr^3 - \hat{k} \int_{\mathbb{R}^3} j\psi(\bar{r}) |\bar{r}\rangle \, dr^3) \tag{14}$$

$$= \int_{\mathbb{R}^3} jk\psi(\bar{r}) |\bar{r}\rangle \, dr^3 - \int_{\mathbb{R}^3} kj\psi(\bar{r}) |\bar{r}\rangle \, dr^3 = 0 \tag{15}$$

Similarly, the action of $[\hat{p}_j, \hat{p}_k]$ on the states as follow:

$$[\hat{p}_j, \hat{p}_k] |\psi\rangle = -\hbar^2 \left(\frac{\partial}{\partial j}\frac{\partial}{\partial k} - \frac{\partial}{\partial k}\frac{\partial}{\partial j} \right) \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle \, dr^3 \tag{16}$$

$$= -\hbar^2 \left(\frac{\partial}{\partial j} \int_{\mathbb{R}^3} \frac{\partial \psi}{\partial k} |\bar{r}\rangle \, dr^3 - \frac{\partial}{\partial k} \int_{\mathbb{R}^3} \frac{\partial \psi}{\partial j} |\bar{r}\rangle \, dr^3 \right) \tag{17}$$

$$= -\hbar^2 \left(\int_{\mathbb{R}^3} \frac{\partial}{\partial j} \left(\frac{\partial \psi}{\partial k} \right) |\bar{r}\rangle \, dr^3 - \int_{\mathbb{R}^3} \frac{\partial}{\partial k} \left(\frac{\partial \psi}{\partial j} \right) |\bar{r}\rangle \, dr^3 \right) = 0 \tag{18}$$

Which, it's due to the fact that the mixed partial derivatives are equal if the function is smooth / at least twice continuously differentiable.

Now, the action of $[\hat{j}, \hat{p}_k]$ is as follow:

$$[\hat{j}, \hat{p}_k] |\psi\rangle = -i\hbar \left(\hat{j}\frac{\partial}{\partial k} - \frac{\partial}{\partial k}\hat{j} \right) \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle \, dr^3 \tag{19}$$

$$= -i\hbar \left(\hat{j} \int_{\mathbb{R}^3} \frac{\partial \psi}{\partial k} |\bar{r}\rangle \, dr^3 - \frac{\partial}{\partial k} \int_{\mathbb{R}^3} j\psi(\bar{r}) |\bar{r}\rangle \, dr^3 \right) \tag{20}$$

$$= -i\hbar \left(\int_{\mathbb{R}^3} j\frac{\partial \psi}{\partial k} |\bar{r}\rangle \, dr^3 - \int_{\mathbb{R}^3} \left(\frac{\partial j}{\partial k}\psi(\bar{r}) + j\frac{\partial \psi}{\partial k} \right) |\bar{r}\rangle \, dr^3 \right) \tag{21}$$

$$= -i\hbar \left(\int_{\mathbb{R}^3} j\frac{\partial \psi}{\partial k} |\bar{r}\rangle \, dr^3 - \delta_{jk} \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle \, dr^3 - \int_{\mathbb{R}^3} j\frac{\partial \psi}{\partial k} |\bar{r}\rangle \, dr^3 \right) \tag{22}$$

$$= i\hbar\delta_{jk} |\psi\rangle \tag{23}$$

So, as a result, we get:

$$[\hat{x}, \hat{y}] = 0, \quad [\hat{x}, \hat{p}_y] = 0, \quad [\hat{x}, \hat{p}_x] = i\hbar, \quad [\hat{p}_y, \hat{p}_z] = 0 \tag{24}$$

(b) If limit to one dimension (pick $j \in \{x, y, z\}$), then Ehrenfest's theorem in one dimension guarantees that $\frac{d}{dt}\langle j\rangle = \frac{1}{m}\langle p_j\rangle$ and $\frac{d}{dt}\langle p_j\rangle = \left\langle -\frac{\partial V}{\partial j}\right\rangle$. Hence, the three-dimensional version follows, since now if $\mathbf{p}$ and $\mathbf{r}$ are $(\hat{p}_x, \hat{p}_y, \hat{p}_z)$ and $(\hat{x}, \hat{y}, \hat{z})$ respectively, one has:

$$\frac{d}{dt}\langle \mathbf{r}\rangle = \frac{d}{dt}(\langle x\rangle, \langle y\rangle, \langle z\rangle) = \frac{1}{m}(\langle p_x\rangle, \langle p_y\rangle, \langle p_z\rangle) = \frac{1}{m}\langle \mathbf{p}\rangle \tag{25}$$

$$\frac{d}{dt}\langle \mathbf{p}\rangle = \frac{d}{dt}(\langle p_x\rangle, \langle p_y\rangle, \langle p_z\rangle) = \left(\left\langle -\frac{\partial V}{\partial x}\right\rangle, \left\langle -\frac{\partial V}{\partial y}\right\rangle, \left\langle -\frac{\partial V}{\partial z}\right\rangle\right) = \langle -\nabla V\rangle \tag{26}$$

Since $-\nabla V = \left(-\frac{\partial V}{\partial x}, -\frac{\partial V}{\partial y}, -\frac{\partial V}{\partial z}\right)$.

(c) Given $j, k \in \{x, y, z\}$, since $[\hat{j}, \hat{p}_k] = i\hbar\delta_{jk}$ (verified in part (a)), then by the generalized uncertainty principle, one has:

$$\sigma_j \sigma_{p_k} \geq \frac{1}{2}\left|[\hat{j}, \hat{p}_k]\right| = \frac{\hbar}{2}\delta_{jk} \tag{27}$$

Hence, if $j = k$, then $\sigma_j \sigma_(p_j) \geq \frac{\hbar}{2}$, yet if $j \neq k$, one has $\sigma_j \sigma_{p_k} \geq 0$, which imposes no restriction.

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Question 5. ***Cubical well*** *(Griffiths 4.2, same in both editions)*
Use separation of variables in cartesian coordinates to solve the infinite cubical well (or "particle in a box"):

$$V(x,y,z) = \begin{cases} 0 & x,y,z \in [0,a] \\ \infty & \textit{otherwise} \end{cases}$$

(a) Find the stationary states, and the corresponding energies.

(b) Call the distinct energies $E_1, E_2, E_3, \ldots$ in order of increasing energy. Find E_1, E_2, E_3, E_4, E_5, and E_6. Determine their degeneracies (that is, the number of different states that share the same energy).

(c) What is the degeneracy of E_{14}, and why is this case interesting?

Proof.

(a) Suppose the wavefunction $\psi(\bar{r}) = \psi_x(x)\psi_y(y)\psi_z(z)$ (which, since $\psi \neq 0$, for some open sets all three functions must be nonzero), then the time-independent Schródinger equation reduces to the following:

$$E\psi_x(x)\psi_y(y)\psi_z(z) = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_x(x)\psi_y(y)\psi_z(z) \tag{28}$$

$$= -\frac{\hbar^2}{2m}\left(\psi_y\psi_z\frac{d^2\psi_x}{dx^2} + \psi_x\psi_z\frac{d^2\psi_y}{dy^2} + \psi_x\psi_y\frac{d^2\psi_z}{dz^2}\right) \tag{29}$$

In particular, on the open subset of the box $[0,a]^3$ where all three functions are nonzero, one gets the following differential equation (by dividing ψ for all the equations):

$$E = -\frac{\hbar^2}{2m}\left(\frac{1}{\psi_x}\frac{\partial^2\psi_x}{\partial x^2} + \frac{1}{\psi_y}\frac{\partial^2\psi_y}{\partial y^2} + \frac{1}{\psi_z}\frac{\partial^2\psi_z}{\partial z^2}\right) \tag{30}$$

Notice that each of the term $\frac{1}{\psi_u}\frac{\partial^2\psi_u}{\partial u^2}$ is only dependent on u, so for the sum of three such terms to be a constant (each dependent only on one distinct variable), each term must be a constant. Hence, there exists $E_u \in \mathbb{R}$, such that $E_u = -\frac{\hbar^2}{2m\psi_u}\frac{\partial^2\psi_u}{\partial u^2}$ (and as a result, $E = E_x + E_y + E_z$).

Which, this implies $E_u\psi_u = -\frac{\hbar^2}{2m}\frac{\partial^2\psi_u}{\partial u^2}$ on the interval $u \in [0,a]$, showing that each ψ_u satisfies the one-dimensional infinite square well Hamiltonian.

Therefore, given any $n_x, n_y, n_z \in \mathbb{N}\backslash\{0\}$, one has the stationary state $\psi_{n_x,n_y,n_z} = \left(\frac{2}{a}\right)^{\frac{3}{2}} \sin\left(\frac{n_x\pi}{a}x\right)\sin\left(\frac{n_y\pi}{a}y\right)\sin\left(\frac{n_z\pi}{a}z\right)$, with energy $E_{n_x,n_y,n_z} = E_{n_x} + E_{n_y} + E_{n_z} = \frac{(n_x^2+n_y^2+n_z^2)\pi^2\hbar^2}{2ma^2}$.

(b) As seen in part (a), the energy of the stationary state must be the nth infinite square well energy, with n being sum of three positive integer squares. Which, the first 14 such positive integers are given as below:

- $3 = 1 + 1 + 1$.
- $6 = 1 + 1 + 2^2$.
- $9 = 1 + 2^2 + 2^2$.

- $11 = 1 + 1 + 3^2$.
- $12 = 2^2 + 2^2 + 2^2$.
- $14 = 1 + 2^2 + 3^2$.
- $17 = 4^2 + 4^2 + 9^2$
- $18 = 1 + 1 + 4^2$
- $19 = 1 + 3^2 + 3^2$.
- $21 = 1 + 2^2 + 4^2$.
- $22 = 2^2 + 3^2 + 3^2$.
- $24 = 2^2 + 2^2 + 4^2$.
- $26 = 1 + 3^2 + 4^2$.
- $27 = 1 + 1 + 5^2 = 3^2 + 3^2 + 3^2$.

Which, consider the degeneracies (which are the distinct 3-tuples with the three positive integers mentioned), one has the first 6 energies (and their degeneracies) as follow:

- $E_1 = \frac{3\pi^2\hbar^2}{2ma^2}$, with $d(1) = 1$ (ground state energy).
- $E_2 = \frac{6\pi^2\hbar^2}{2ma^2}$, with $d(2) = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3$ (need to choose which coordinate does 2 correspond to).
- $E_3 = \frac{9\pi^2\hbar^2}{2ma^2}$, with $d(3) = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3$ (need to choose which coordinate does 1 correspond to).
- $E_4 = \frac{11\pi^2\hbar^2}{2ma^2}$, with $d(4) = 3$ (again, choosing which coordinate does 3 correspond to).
- $E_5 = \frac{12\pi^2\hbar^2}{2ma^2}$, with $d(5) = 1$ (since all three coordinates correspond to 2).
- $E_6 = \frac{14\pi^2\hbar^2}{2ma^2}$, with $d(6) = 6$ (since it's all the possible permutation of 3 distinct positive integers).

(c) From the list above, one can see the 14th sum of three positive integer square, is $27 = 1 + 1 + 5^2 = 3^2 + 3^2 + 3^2$. So one can see there are more than one way of expressing it as sum of three positive integer squares. Then, the degeneracy is in fact $d(14) = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + 1 = 4$ (one corresponds to $1 + 1 + 5^2$, the other corresponds to $3^2 + 3^2 + 3^2$).

$\square$

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Question 6. ***Practice with Spherical Harmonics*** *(Griffiths 4.4 in 3rd edition, 4.3 in 2nd edition)*
Show that

$$\Theta(\theta) = A\ln[\tan(\theta/2)]$$

satisfies the Θ *equation for* $\ell = m = 0$. *This is the unacceptable "second solution" - what's wrong with it?*
Note: the equation is of the form

$$\sin(\theta)\frac{d}{d\theta}\left(\sin(\theta)\frac{d\Theta}{d\theta}\right) + [\ell(\ell+1)\sin^2(\theta) - m^2]\Theta = 0$$

Proof. Given $\ell = m = 0$, the given equation reduces to:

$$\sin(\theta)\frac{d}{d\theta}\left(\sin(\theta)\frac{d\Theta}{d\theta}\right) = 0 \tag{31}$$

Hence, one gets $\sin(\theta)\frac{d\Theta}{d\theta} = A_1$ for some constant $A_1 \in \mathbb{C}$.

Now, since this implies $\frac{d\Theta}{d\theta} = \frac{A_1}{\sin(\theta)} = A_1\csc(\theta)$, then one gets the following:

$$\Theta = \int A_1 \csc(\theta) d\theta = -A_1 \ln\left|\tan\left(\frac{\theta}{2}\right)\right| + A_2 \tag{32}$$

Which, for $\theta \in (0,\pi)$, $\tan(\theta/2) > 0$, hence $\Theta = A\ln[\tan(\theta/2)] + B$ for some $A, B \in \mathbb{C}$.

However, this is not an acceptable solution, since the integration of $|\Theta|^2$ over $\theta \in (0,\pi)$ in fact diverges, showing that Θ is not normalizable. $\square$